Multiview Multilevel Robust Deep Subspace Clustering for Hyperspectral Band Selection

Dongkai Yan, Xiaodi Shang, *Member, IEEE*, Xiaopeng Wang, Jiahua Zhang, and Xudong Sun, *Member, IEEE*

***Abstract*—Subspace clustering (SC)-based band selection (BS) have achieved significant progress. However, traditional approaches focus solely on the linear relationships among bands, neglecting important nonlinear information. Moreover, single-view data inherently limits the exploitation of complementary latent features across different views. To address the above issues, this article proposes a novel multiview multi-view multilevel robust DSC (MRDSC). First, we construct spectral, spatial, and texture views to provide rich and complementary feature representations for SC. Further, a multilevel deep SC network is employed to deeply capture both global and local nonlinear relationships within the self-representation coefficient matrix of each view, while Robust Principal Component Analysis is applied to obtain a clean coefficient matrix. Finally, a self-aligned contrastive (SAC) learning mechanism is introduced to enhance the consistency of positive samples and the discriminability of negative samples, effectively integrating complementary information across views and reinforcing feature consistency. Experimental results on three public datasets demonstrate that the proposed method exhibits high stability and effectiveness.**

***Index Terms*—Band selection (BS), contrastive learning (CL), deep subspace clustering (DSC), multiview learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide rich spectral information to distinguish differences in land cover materials, and have been widely used in applications such as vegetation estimation, environmental monitoring, forest protection, and ocean exploration [1]. However, the high dimensionality of HSI not only increases the complexity of data processing but also leads to inevitable band redundancy and noise, significantly degrading the performance of subsequent analysis [2]. Therefore, dimensionality reduction is essential to alleviate the "curse of dimensionality" [3].

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Feature extraction (FE) and band selection (BS) are two common methods for HSI dimensionality reduction [4]. Both seek to reduce computational complexity while maximizing information retention. Specifically, FE projects the original high-dimensional data into a new, low-dimensional feature space [5], but this often results in the loss of the physical meaning of the original bands. In contrast, BS reduce dimensionality by selecting the most important and representative bands to form a data subset [6], thereby preserving the intrinsic spatial and spectral features of the HSI. Therefore, this article focuses on the study of BS methods [7].

Existing BS methods can be categorized into ranking-based, searching-based, deep learning (DL)-based, sparsity-based, and clustering-based approaches. Ranking-based methods sort original bands according to specific criteria and select the top-ranked ones to form a band subset. Typical examples include maximum-variance principal component analysis (MVPCA) [8] and similarity-based ranking method (SR-SSIM) [9]. Searching-based methods formulate the BS problem as an optimization task with a predefined objective function and employ specific search strategies to identify the optimal band subset. Representative methods include the unsupervised BS based on information theory and a modified discrete gravitational search algorithm (MIMR-DGSA) [10], and the higher order joint volume gradient (HOJVG) [2]. Sparsity-based methods such as sparsity-based BS (SpaBS) [11] and joint spatial and spectral graph-based consistent self-representation (JSSGCSR) [12], employ sparse representation to capture linear correlations among bands and obtain representative bands. DL-based BS methods leverage the strong fitting capabilities of neural networks to reveal complex interband dependencies in HSI and select key bands accordingly. Examples include the dual-attention reconstruction network for BS (DARecNet-BS) [13], the spectral–spatial cross-dimensional attention network (SSANet-BS) [14] and an end-to-end spectral–spatial hypergraph convolutional network (S^2HGC) [15]. Clustering-based methods, represented by the dual-clustering-based BS [16], the optimal clustering framework (OCF) [17], and the block diagonal representation learning algorithm (BDRLA) [18] apply clustering algorithms to group highly similar bands, thereby achieving the selection of representative bands from each group.

In recent years, self-representation-based subspace clustering (SC) methods have gained significant attention due to their effectiveness in modeling the linear relationships

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among bands. These methods construct a dictionary from the data itself and encode the interband relationships by solving for a self-representation coefficient matrix, thereby revealing the underlying affinity structure among bands. For example, the improved sparse subspace clustering (ISSC) model [19] introduces sparse representation into SC by employing sparse subspace clustering (SSC) to capture band similarities. However, in SSC, each band can only be represented by other bands within the same subspace, which limits the coefficient matrix in capturing global structural information [20] and degrades its performance in BS.

To address these issues, low-rank constraints have been introduced into SC, such as the nonnegative low-rank representation (NLRR) [21], the Laplacian-regularized low-rank SC (LLRSC) [22], and the fast and latent low-rank SC (FLLRSC) [23]. These methods significantly enhance the robustness of BS by improving the global structure representation. Furthermore, considering the spatial feature differences in HSI, approaches such as graph regularized spatial–spectral SC (GRSC) [24], region-aware hierarchical latent feature representation learning-guided clustering (HLFC) [25], and dual-view structural similarity SC (DVS^{3}C) [26] utilize entropy rate superpixel segmentation (ERS) to generate different regions and extract low-dimensional latent features from heterogeneous regions. Additionally, tensor-based SC model (TSC) [27] and tensorial global–local graph self-representation (TGSR) [28], utilize tensors to further explore band correlations while simultaneously preserving the local and global geometric structures.

However, traditional SC methods are generally limited to modeling the linear relationships among bands, whereas DL can capture more complex nonlinear dependencies and has therefore received growing attention [29]. Representative techniques include autoencoder (AE) [30], convolutional neural networks (CNNs) [31], and attention mechanisms [32]. Recently, DL has been integrated with SC through deep subspace clustering (DSC) to explore nonlinear spectral relationships. For instance, Zeng et al. [33] integrated the SC into a convolutional AE to fully extract the nonlinear relationships between spectra. Considering that sparse self-representation can reduce spectral redundancy, Das et al. [34] proposed a sparsity regularized DSC for multicriterion-based hyperspectral BS, combining structural similarity and KL divergence to select representative bands from each cluster. Furthermore, Wang et al. [35] considered the multilevel spectral information of HSI and proposed a self-supervised deep multilevel representation learning fusion-based maximum entropy SC (MLRLFMESC). It integrates the multilevel spectral–spatial information from a convolutional AE to obtain a more comprehensive coefficient matrix.

Although existing deep subspace clustering-based BS (DSC-BS) methods can capture nonlinear dependencies among bands, several challenges remain. First, most methods exploit only the dependencies within the original spectral view, while overlooking cross-view associations. Second, when constructing the self-representation coefficient matrix, conventional DSC-BS methods typically rely solely on the latent features from the final decoder layer, thereby underutilizing the hierarchical representations of the encoder. Finally, effectively leveraging the consistency and complementarity across multiple views remains a critical yet unresolved issue for DSC-BS.

In summary, this article proposes a multiview multilevel robust DSC (MRDSC) model. First, building upon the original spectral view, spatial and texture view are constructed using ERS and Gabor filter techniques, respectively, aiming to jointly mine the multidimensional latent information embedded in HSI from multiple views. Second, an end-to-end multilevel robust DSC framework is designed. Through iterative optimization, it synchronously extracts local and global nonlinear relationships from the latent features at each level of each view and derives a clean self-representation matrix. Furthermore, MRDSC introduces a self-aligned contrastive (SAC) learning mechanism that strengthens the consistency of positive samples and the discriminability of negative samples between each view and the consensus matrix, thereby fully exploiting complementary and consistent information across views. Finally, MRDSC performs spectral clustering, and selects the optimal band subset. Specifically, the main contributions of this work are summarized as follows.

1) This article proposes a novel BS method, MRDSC, which constructs spectral, spatial, and texture views to synergistically explore multidimensional information in HSI, achieving complementary collaborative learning across the three views.
2) A multilevel robust DSC framework is developed, which captures both local and global nonlinear relationships within each level of every view through end-to-end iterative optimization.
3) A self-aligned CL mechanism is introduced to enhance the alignment consistency of positive samples while increasing the discriminability of negative samples. This process effectively promotes the synergistic fusion of consistent and complementary information among the multiple views.

The remainder of this article is organized as follows. Section II reviews the related work on multiview clustering (MVC), multiview DSC and CL. Section III details the proposed MRDSC method for unsupervised hyperspectral BS. Section IV reports the experimental results, and Section V discusses the findings. Finally, Section VI concludes the article.

II. Related Work

In this section, we review the research relevant to the proposed method, with a focus on MVC, multiview deep SC and CL.

A. Multiview Clustering

MVC aims to improve clustering performance by fusing complementary information from multiple views. This approach constructs a more discriminative feature representation to reveal the intrinsic affinity relationships in the data, thereby overcoming the limitations of a single view. Zhu et al. [36] proposed a novel latent heterogeneous graph network

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(LHGN) for incomplete multiview learning. It learns a unified latent representation to fully integrate information from multiple incomplete views, and implicitly achieves a balance between inter-view consistency and complementarity. Liu et al. [37] proposed a novel GCN-based deep multiview graph clustering network with weighting mechanism and collaborative training (DMVGC). They designed a collaborative architecture of view-specific encoders and a unified decoder to dynamically balance view diversity and consistency. In addition, integrated heterogeneous graph attention (IHGAT) network [38] for incomplete multimodal clustering builds a set of integrated heterogeneous graphs by fusing sample similarities with the modality-specific availability graphs. Then it employs a graph attention network to aggregate multiview information for learning robust representations.

In recent studies, numerous scholars have constructed multiview data from original HSI, with the aim of further enhancing the performance of relevant downstream tasks. Li et al. [39] proposed a novel subpixel-pixel-superpixel-based multiview active learning (SPS-MAL) method. It generates multiview data by extracting subpixel, pixel, and super-pixel information from the HSI, thereby leveraging diverse and complementary information for better category recognition. Luo et al. [40] proposed the semantic constraint-based spatial–spectral multi view clustering (SCMVC) method, which designs spatial multiview learning to capture distinct features for contrastive clustering. Overall, these studies demonstrate that effectively exploiting complementarity and consistency information across multiple views is crucial for enhancing feature representation in HSI analysis.

B. Multiview Deep Subspace Clustering

SC techniques have found wide applications in computer vision, such as mechanical fault diagnosis [41] and image processing [42]. With the advancement of DL, DSC has attracted increasing attention from researchers. Typical DSC Networks [43] utilize deep AE to extract low-dimensional latent features from high-dimensional raw data, while employing a fully connected layer to learn the self-representation matrix for a single view. The decoder subsequently reconstructs the data, with the consistency representation learned by minimizing the reconstruction error of the input data. The basic objective function of DSC can be expressed as

$$\min \left\|\mathbf{X}-\hat{\mathbf{X}}\right\|_F^2+\|\mathbf{Z}-\mathbf{ZC}\|_F^2+\lambda\|\mathbf{C}\|_F \tag{1}$$

where $\mathbf{X}$ and $\hat{\mathbf{X}}$ denote the input and reconstructed data. $\mathbf{Z}$ is the latent feature extracted from the last layer of the encoder, and $\mathbf{C}$ is the self-expression coefficient matrix that characterizes the relationships among latent features. $\|\cdot\|_F$ is Frobenius norm.

In the context of HSI processing, Han et al. [44] applied DSC to the task of HSI processing. They proposed the deep low-rank graph convolutional subspace clustering (DLRGCSC) framework. It utilizes 1-D and 2-D graph convolutional AEs to simultaneously extract HSI features and constructs a low-rank constrained self-expression layer, significantly enhancing the discriminability of the joint features.

In practical scenarios, multiview data not only contains cross-view consistent information but also provides complementary and diverse features, significantly enhancing the discriminative performance of clustering. Therefore, some multiview DSC (MVDSC) methods have been proposed. Self-guided deep multiview subspace clustering via consensus affinity regularization (SDMSC) [45] significantly improves clustering discriminative performance by constraining the data affinity consistency between multiple views and between embedding spaces. Wang et al. [46] proposed multiscale deep multiview subspace clustering (MDMVSC). It jointly learns global and local latent features to generate a higher-quality and more comprehensive self-representation matrix for each view. In addition, Zhu et al. [47] learn multi-view self-representations in an end-to-end manner. Through explicitly decoupled diversity and commonality networks, they embed multiview relationships into the feature learning process, improving the clustering discriminative performance. The model of MVDSC is as follows:

$$\min \sum_{v=1}^{V}\left\|\mathbf{X}^v-\hat{\mathbf{X}}^v\right\|_F^2+\sum_{v=1}^{V}\|\mathbf{Z}^v-\mathbf{Z}^v\mathbf{C}^v\|_F^2+\sum_{v=1}^{V}\lambda\|\mathbf{C}^v\|_F. \tag{2}$$

Here, V denotes the number of views. $\mathbf{X}^v$ and $\hat{\mathbf{X}}^v$ represent the original and reconstructed data of vth view, respectively, while $\mathbf{Z}^v$ and $\mathbf{C}^v$ are the corresponding latent features and self-representation coefficient matrix of $\mathbf{X}^v$.

C. Contrastive Learning

CL is an unsupervised technique that learns high-quality data representations by constructing positive and negative sample pairs. Its core principle involves mapping raw samples into a low-dimensional embedding space, where a similarity metric quantifies the representation differences between sample pairs. CL constrains the representation distance of positive pairs to be minimized while enforcing the representation distance of negative pairs to be maximized, thereby yielding generalizable and discriminative information.

In recent years, CL has been widely applied in the field of multiview processing. For example, Fei et al. [48] propose a new deep multiview contrastive clustering method via graph structure awareness (DMvCGSA). This method simultaneously performs instance-level and cluster-level CL to achieve collaborative representation of multiview samples. Yu et al. [49] propose a novel multiview DSC method based on CL and Cauchy–Schwarz (CS) divergence, utilizing an extension of the multiple positive and negative pair diffusion method to enhance CL. Guan et al. [50] proposed a contrastive multiview subspace clustering of HSI based on graph convolutional networks (CMSCGC), which aggregates multiview information through graph convolutional networks while incorporating a CL mechanism to enhance positive sample consistency. Furthermore, several researchers have introduced CL into other HSI processing, such as classification [51] and target detection [52].

III. MRDSC

This section provides a detailed explanation of the proposed MRDSC. The framework is illustrated in Fig. 1. The MRDSC

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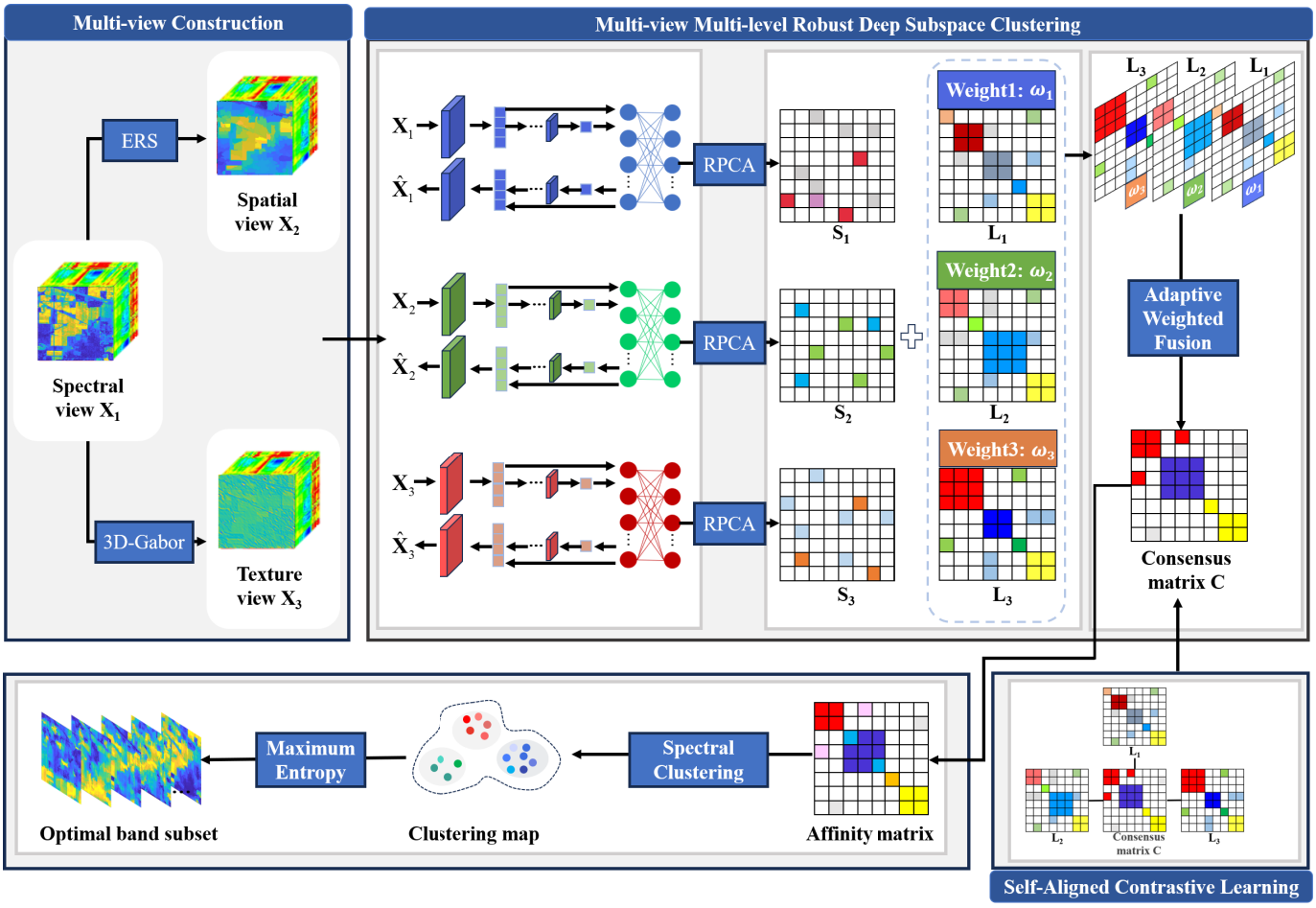


Fig. 1. Flowchart of MRDSC.

is mainly composed of three modules: multiview construction, MRDSC, and self-aligned contrastive (SAC) learning. The specific description of the MRDSC is as follows.

A. Multiview Construction

HSI contains rich and diverse information, yet a single view is often insufficient to comprehensively represent their complex features. Building upon the HSI (spectral view), this article utilizes the ERS and Gabor filters to construct spatial and texture views, extracting image features with diversity and complementarity from different dimensions.

1) Spectral View $\mathbf{X}^1$*:* HSI typically consists of tens to hundreds of continuous and narrow bands. These dense bands hold the fine spectral responses of ground objects, providing a crucial basis for distinguishing targets with similar spatial morphologies but different material compositions. Formally, the original HSI can be represented as $\mathbf{X} = \{\mathbf{x}_i | \mathbf{x}_i \in \mathbb{R}^{N\times 1}\}_{i=1}^{B}$, where N donates the number of pixels in each band $\mathbf{x}_i$ and B is the spectral dimension. We refer to $\mathbf{X}$ as the spectral view of HSI.

2) Spatial View $\mathbf{X}^2$*:* Different regions in HSI typically correspond to distinct land-cover types, while adjacent pixels within the same homogeneous region tend to exhibit high spectral similarity. Consequently, fully exploiting the spatial structural information can provide effective assistance for selecting sensitive bands tailored to particular ground types.

To effectively utilize the spatial information of HSI, this article constructs a spatial view using ERS. By aggregating spatially adjacent pixels with similar spectral features into homogeneous superpixels, this method extracts and represents the spatial structural information of the image. Specifically, ERS is applied to the first principal component $\mathbf{M}$ of the HSI, partitioning the image into K homogeneous regions

$$\mathbf{M} = \bigcup_{i=1}^{K} \mathcal{R}_i \quad \text{s.t. } \mathcal{R}_i \cap \mathcal{R}_j = \emptyset \quad (\forall i, j, i \neq j) \tag{3}$$

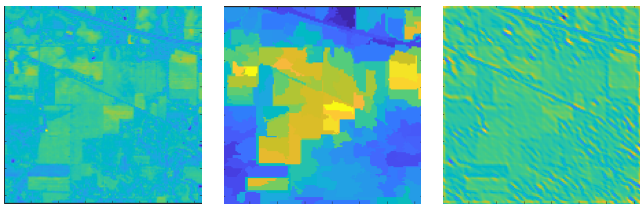
where $\mathcal{R}_i$ denotes the ith segmented region. Owing to the high spectral similarity within each superpixel, we replace the spectrum of each pixel $\mathbf{y}_i \in \mathbb{R}^{B\times 1}$ in region $\mathcal{R}_k$ with the average spectrum $\bar{\mathbf{y}}_k$, thus constructing the spatial view $\mathbf{X}^2$

$$\bar{\mathbf{y}}_k = \frac{1}{N_k} \sum_{\mathbf{y}_i \in \mathcal{R}_k} \mathbf{y}_i. \tag{4}$$

Here, N_k denotes the number of pixels in the kth superpixel $\mathcal{R}_k$.

3) Texture View $\mathbf{X}^3$*:* Texture information can capture the fine-scale edge details in HSI land covers. To exploit this property, we construct a texture view and combine it with the

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(a) (b) (c)

Fig. 2. Different views at the 100th spectral band of Indian Pines dataset. (a) Spectral view. (b) Spatial view. (c) Texture view.

spatial view to enhance material discrimination. The Gabor filter is a commonly used technique for textural information extraction. By designing the filters with specific frequencies and orientations, more potential structural textural features can be extracted, showing excellent performance in HSI data analysis. Specifically, the 3-D-Gabor kernel is designed as follows:

$$g(x,y,z) = \hat{g}(x,y,z)\, e^{j2\pi(\mu_0 x + v_0 y + w_0 z)}. \tag{5}$$

In (5), $\hat{g}(x,y,z)$ donates the 3D Gaussian envelope in the (x,y,z) domain. The terms $u_0 = \omega \sin\theta\cos\phi$, $v_0 = \omega \sin\theta \sin\phi$, $w_0 = \omega\cos\theta$ are the frequency components along the x, y, and z directions, respectively. ω is the frequency parameter, and $\{\theta, \phi\}$ define the orientation. In this article, ω and orientation $\{\theta,\phi\}$ are set to $\omega \in \{1/4, 1/12\}$ and $\{\theta,\phi\} \in \{0, \pi/4, \pi/2, 3\pi/4\}$, which are commonly used values in 3D-Gabor filtering techniques [53].

Finally, the normalized product of Shannon entropy (SE) and the Tenengrad gradient (TG) is employed as the evaluation metric to determine the optimal parameter combination. Specifically, for each parameter combination, the average SE and TG values across all bands of each dataset are calculated. The combination yielding the highest normalized product is selected as the optimal parameter and subsequently applied to construct the texture view. SE quantifies the information richness of the texture view, while TG measures the edge sharpness of the image texture. The definitions of these two metrics are given below

$$\mathrm{SE}(\mathbf{x}_i) = -\sum_{k=0}^{n} p_k \log_2 p_k \tag{6}$$

$$\mathrm{TG}(x,y) = \sum_x \sum_y \left(G_x^2(x,y) + G_y^2(x,y)\right) \tag{7}$$

where $\mathbf{x}_i$ donates the ith band, n is the maximum pixel value in $\mathbf{x}_i$, and p_k is the ratio of pixels with value k to the total number of pixels. $G_x(x,y)$ and $G_y(x,y)$ are the horizontal and vertical Sobel operator at pixel (x,y). Finally, the optimal parameter combinations $\{\omega,\theta,\phi\}$ for Indian Pines, Pavia University (PU), and Salinas datasets used in the experiments are $\{1/4, \pi/4, 0\}$, $\{1/4, \pi/2, 3\pi/4\}$, and $\{1/4, \pi/4, 0\}$, respectively.

Fig. 2 illustrates the spectral, spatial, and texture views at the 100th band. These three views provide complementary information from different perspectives. Specifically, the spectral view presents the most faithful distribution of land-cover materials, the spatial view captures the structural characteristics of the image, and the texture view reveals edge details and textural features of land-cover materials. Such complementarity effectively enhances the performance of subsequent tasks.

B. Multiview Multilevel Robust Deep Subspace Clusering

The MRDSC module is primarily composed of four parts: an encoder, a view-specific self-representation layer, a decoder, and an adaptive weight fusion layer, respectively.

1) Encoder: For a given view $\mathbf{X}^v$, $1 \le v \le 3$, the encoder is composed of L independent convolutional neural networks, $E^v = \{E_1^v, E_2^v, \ldots, E_L^v\}$. It is employed to learn the multilevel latent features $\mathbf{Z}^v = \{\mathbf{Z}_1^v, \mathbf{Z}_2^v, \ldots, \mathbf{Z}_L^v\}$, which are described as follows:

$$\mathbf{Z}_l^v = E_l^v\left(\mathbf{Z}_{l-1}^v; \theta_{E_l^v}\right) \tag{8}$$

where $l = 1, 2, \ldots, L$ represents the index of encoder layer, and $\theta_{E_l^v}$ is the parameter of the lth layer encoder E_l^v. In the model, the input to the encoder is the original view data.

2) View-Specific Self-Representation Layer: To effectively exploit the local–global relationships of each view, a linear self-representation layer is constructed. Based on the multilevel latent features $\{\mathbf{Z}_l^v\}_{l=1}^L$, a view-specific self-representation coefficient matrix $\mathbf{A}^v \in \mathbb{R}^{B\times B}$ is learned for each view $\mathbf{X}^v$. This layer performs a purely linear transformation without nonlinear activation functions or bias terms. Its mathematical expression is given as follows:

$$\{\mathbf{Z}_l^v\}_{l=1}^L = \{\mathbf{Z}_l^v\}_{l=1}^L \mathbf{A}^v. \tag{9}$$

However, $\mathbf{A}^v$ is inevitably affected by factors such as noise, which impacts subsequent BS performance. To improve the robustness of the coefficient matrix, we employ RPCA to decompose $\mathbf{A}^v$ into a clean coefficient matrix $\mathbf{L}^v$ and a noise matrix $\mathbf{S}^v$. Then, (9) can be rewritten as

$$\{\mathbf{Z}_l^v\}_{l=1}^L = \{\mathbf{Z}_l^v\}_{l=1}^L (\mathbf{L}^v + \mathbf{S}^v). \tag{10}$$

3) Decoder: The decoder $D^v = \{D_1^v, D_2^v, \ldots, D_L^v\}$ employs a transposed convolutional network, symmetric to the encoder, to reconstruct the multiview features layer by layer. Specifically, the reconstructed vector from the last layer of the encoder serves as the input to the first layer of the decoder, i.e. $\hat{\mathbf{Z}}_1^v = \mathbf{Z}_1^v \mathbf{C}^v$. The output of each decoder layer is then used as the input to the subsequent layer

$$\hat{\mathbf{Z}}_{l+1}^v = D_l^v\left(\hat{\mathbf{Z}}_l^v \mathbf{L}^v; \theta_{D_l^v}\right) \tag{11}$$

where D_l^v denotes the lth decoder, $\hat{\mathbf{Z}}_{l+1}^v$ represents the output of D_l^v for the vth view, $\mathbf{L}^v$ is the clean coefficient matrix for the vth view, and $\theta_{D_l^v}$ is the parameter of the decoder D_l^v. When $l = L$, the decoder output corresponds to the reconstructed data $\hat{\mathbf{X}}^v$. This framework utilizes a multilevel robust DSC structure, jointly optimizing view reconstruction and subspace representation to capture the local–global correlations among spectral bands.

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4) Adaptive Weight Fusion: Due to the heterogeneity of multiview data, significant differences may exist among views. Therefore, directly merging them into a unified consensus matrix with equal contribution is not an optimal strategy. To overcome this, MRDSC adaptively assigns weights guided by the reconstruction error of each view. This strategy not only preserves the unique discriminative features of the spectral, spatial, and texture views but also enabling a more reasonable fusion of information. The weight ω^v is defined as follows:

$$\omega^v = \frac{1}{2\|\mathbf{X}^v - \hat{\mathbf{X}}^v\|_F^2}. \tag{12}$$

However, the scales of reconstruction errors differ across various views. Directly using the original reconstruction errors to assign weights may lead to excessively large or small values, potentially neglecting the contributions of certain views. To avoid the aforementioned problem, a normalized weighting mechanism is proposed as follows:

$$\bar{\omega}^v = \frac{\omega^v}{\sum_{v=1}^{V} \omega^v}. \tag{13}$$

Such weight normalization allows for a more rational assessment of contributions from different views while ensuring the consensus matrix $\mathbf{C}$ remains balanced across views. The final $\mathbf{C}$ is defined as

$$\mathbf{C} = \sum_{v=1}^{V} \bar{\omega}^v \mathbf{L}^v. \tag{14}$$

In MRDSC, the self-representation coefficient matrix is learned by enforcing the similarity between the original image and its reconstruction, as well as between the multilevel latent features and their reconstructed counterparts. The corresponding loss function, ℓ_{MVDSC}, is defined as

$$\ell_{\text{MVDSC}} = \sum_{v=1}^{V} \left\|\mathbf{X}^v - \hat{\mathbf{X}}^v\right\|_F^2 + \sum_{v=1}^{V}\sum_{l=1}^{L} \left\|\mathbf{Z}_l^v - \mathbf{Z}_l^v\left(\mathbf{L}^v + \mathbf{S}^v\right)\right\|_F^2 + \sum_{v=1}^{V} \left(\lambda_1 \|\mathbf{L}^v\|_* + \lambda_2 \|\mathbf{S}^v\|_1\right). \tag{15}$$

Here, λ_1 and λ_2 are regularization parameters, $\|\cdot\|_*$ is the nuclear norm, and $\|\cdot\|_1$ denotes the ℓ_1-norm.

C. Self-Aligned CL

Although the adaptive weight fusion mechanism can effectively capture the coefficient correlations within each view, the deep encoders for three views generate their self-representation coefficient matrices independently. This process can make it difficult to fully exploit the complementary and consistent information among the views. More critically, a synergistic optimization mechanism between the clean coefficient matrix $\mathbf{L}^v$ of each view and $\mathbf{C}$ is lacking. This absence prevents the effective transfer of knowledge across views.

To this end, this article introduces SAC learning. The core principle of CL is to construct positive and negative samples, maximizing the similarity between positive samples while minimizing the similarity between negative samples. In this work, the corresponding nodes $(\mathbf{l}_i^v, \mathbf{c}_i)$ from the matrix $\mathbf{L}^v$ and the consensus matrix $\mathbf{C}$ are treated as positive samples, whereas the remaining different nodes $(\mathbf{l}_i^v, \mathbf{l}_k^v)$ and $(\mathbf{l}_i^v, \mathbf{c}_j)$ are regarded as negative samples. Fig. 3 illustrates the positive and negative pairs. As previously stated, the goal is to push $\mathbf{l}_i^v$ away from negative samples while bringing it closer to positive samples. Accordingly, the function is designed as

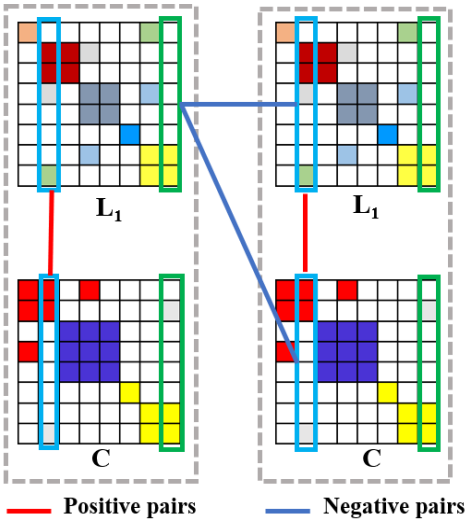


Fig. 3. Illustration of positive and negative pairs.

$$\ell\left(\mathbf{l}_i^v, \mathbf{c}_i\right) = -\log \frac{e^{\left(\frac{\text{sim}\left(\mathbf{l}_i^v, \mathbf{c}_i\right)}{\tau}\right)}}{\sum_{k=1}^{B} e^{\left(\frac{\text{sim}\left(\mathbf{l}_i^v, \mathbf{l}_k^v\right)}{\tau}\right)} + \sum_{j=1}^{B} e^{\left(\frac{\text{sim}\left(\mathbf{l}_i^v, \mathbf{c}_j\right)}{\tau}\right)}} \tag{16}$$

where τ is the temperature parameter, and $\text{sim}(\cdot)$ donates the cosine similarity function. The overall objective of SAC learning is to average the contrastive loss over all positive pairs. Therefore, the SAC loss is

$$\ell_{\text{SAC}} = \frac{1}{2B} \sum_{v=1}^{V} \sum_{i=1}^{B} \left(\ell\left(\mathbf{l}_i^v, \mathbf{c}_i\right) + \ell\left(\mathbf{c}_i, \mathbf{l}_i^v\right)\right). \tag{17}$$

MRDSC effectively captures the consistent and complementary information among multiple views by constraining the distribution of positive and negative sample pairs. This process guides the multiview coefficient matrix representations to align with a shared subspace structure, enhancing cross-view consistency. It is worth noting that direct pairwise CL between views can easily lead to excessive homogenization of the self-representation coefficient matrices, resulting in the loss of complementary information. Therefore, the self-aligned CL strategy addresses this by aligning each view's self-representation matrix $\mathbf{L}^v$ to a global consensus matrix $\mathbf{C}$ instead of enforcing direct constraints between views. This approach effectively extracts cross-view consistency information while maximally preserving the specific complementary information of each view.

In summary, by integrating the multiview multilevel DSC loss with the SAC loss, the overall loss function for the proposed MRDSC is given by

$$\ell_{\text{MRDSC}} = \ell_{\text{MVDSC}} + \lambda_3 \ell_{\text{SAC}}. \tag{18}$$

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Algorithm 1 MRDSC

Input: The HSI dataset $\mathbf{X}$, the number of views V, the number of selected bands n_{BS}, and the regularization parameters λ_1, λ_2, and λ_3.

Output: The optimal band subset.

Set batch size $batch = B$, initial learning rate $lr = 0.001$, and training $epochs = 200$.
Construct spatial view $\mathbf{X}^2$ and texture view $\mathbf{X}^3$ via ERS and 3D-Gabor filter.
for each $epoch$ **do**
    Obtain the multi-level latent features $\{\mathbf{Z}_l^v\}_{l=1}^L$ by Eq. (8).
    Get the clean coefficient matrix $\mathbf{L}^v$ by Eq. (10).
    Obtain the reconstructed data $\hat{\mathbf{X}}^v$ by Eq. (11).
    Update the weight of each view ω^v by Eq. (12).
    Get the consensus matrix $\mathbf{C}$ by Eq. (14).
    Calculate the MVDSC loss ℓ_{MVDSC} and the SAC loss ℓ_{SAC} by Eq. (15) and Eq. (17).
    Jointly optimize the overall loss ℓ_{MRDSC} in Eq. (18) by Adam optimizer.
end for
Construct the affinity matrix $\mathbf{W}$ and performs spectral clustering.
Select bands with the maximum entropy in each cluster.

In (18), λ_3 is the regularization parameter for ℓ_{SAC}. Finally, $\mathbf{C}$ is used to construct an affinity matrix for spectral clustering, and the band with the highest entropy from each cluster is selected to form the optimal band subset. Algorithm 1 details the overall procedure for the proposed MRDSC.

IV. Experiments

In this section, a series of experiments were conducted on three real-world hyperspectral datasets, Indian Pines, PU, and Salinas to evaluate the effectiveness of the proposed MRDSC method. Detailed information on the experimental data, parameter settings, and evaluation metrics is provided. Additionally, the proposed method is compared with eight state-of-the-art BS algorithms, with comparisons focusing on the classification accuracy of the selected band subsets and their performance across different datasets.

A. Datasets

Indian Pines (IP) dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992 over Indian Pines test site in northwestern Indiana. This dataset consists of 220 spectral bands spanning a wavelength range of 0.4–2.5 $^{\mu}$m. The spatial dimension of the dataset is 145 × 145 pixels. It encompasses 16 distinct land-cover classes. The false-color image and ground truth are shown in Fig. 4(a).

The PU dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in 2002 over the Pavia area. This dataset comprises 115 spectral bands with a spatial resolution of 1.3 m, encompassing 9 distinct land-cover classes. The dimension of each spectral image is 610 × 340 pixels. After removing 12 noisy bands, 103 bands were retained for subsequent analysis. The false-color image and ground truth are shown in Fig. 4(b).

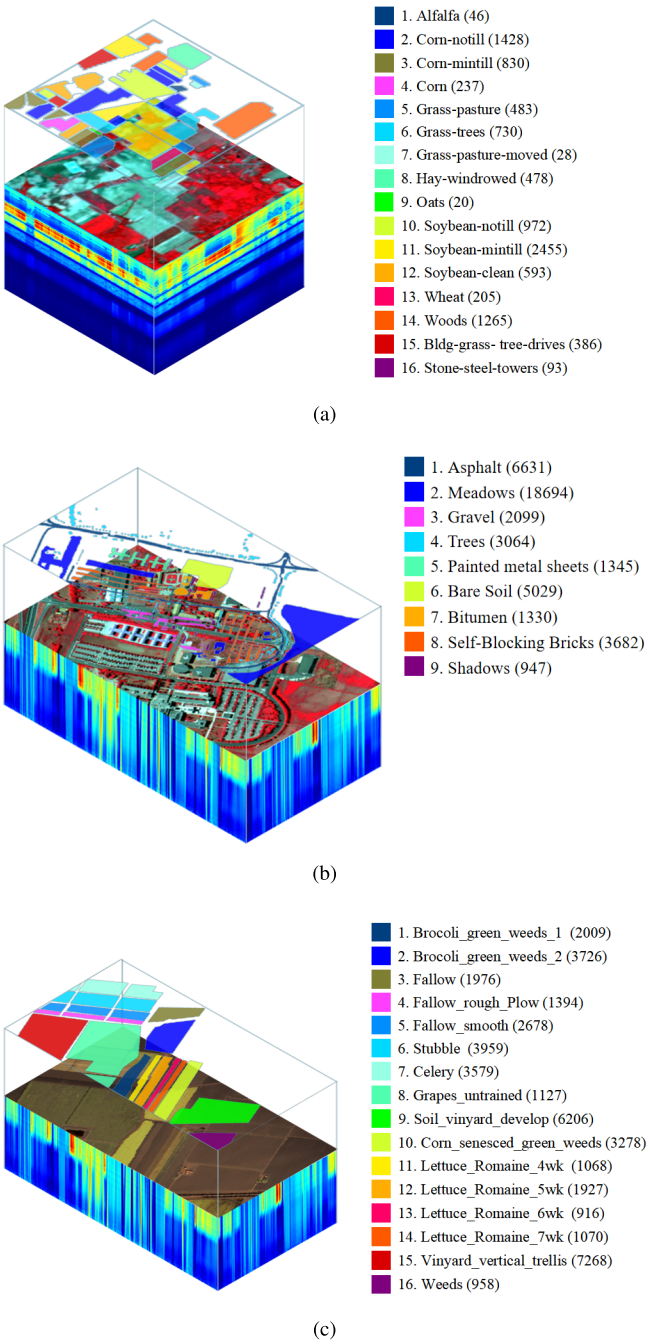


Fig. 4. Datasets used in the experiment. The land cover types and the number of samples for each dataset are indicated. (a) IP. (b) PU. (c) SA.

The Salinas (SA) dataset was collected in 1998 by the AVIRIS sensor over the Salinas Valley, California. This dataset contains 224 spectral bands with a spatial resolution of 3.7 m. It covers 16 distinct land-cover types. The spatial dimension of the dataset is 512 × 217 pixels. The false-color image and ground truth are shown in Fig. 4(c).

B. Experimental Setup

1) Comparison Methods: Eight representative BS methods were employed for comparison, including OCF [17],

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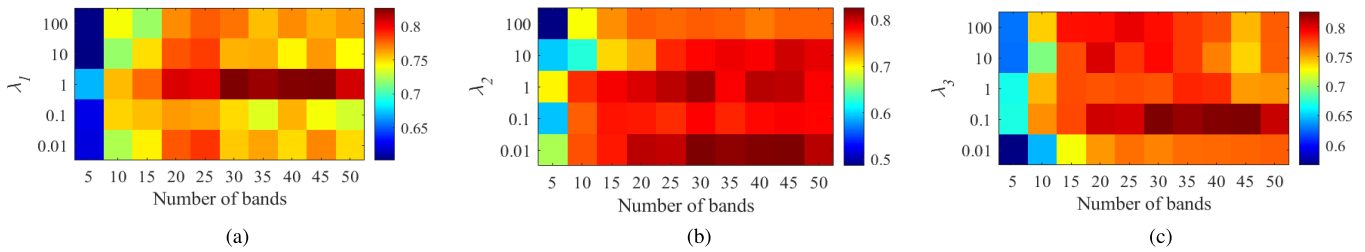


Fig. 5. Sensitivity test of parameters in terms of OA by SVM on IP dataset. (a) λ_1. (b) λ_2. (c) λ_3.

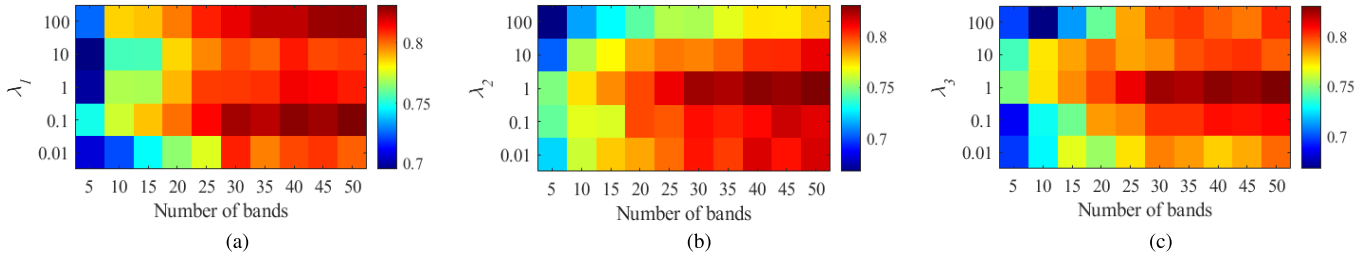


Fig. 6. Sensitivity test of parameters in terms of OA by SVM on PU dataset. (a) λ_1. (b) λ_2. (c) λ_3.

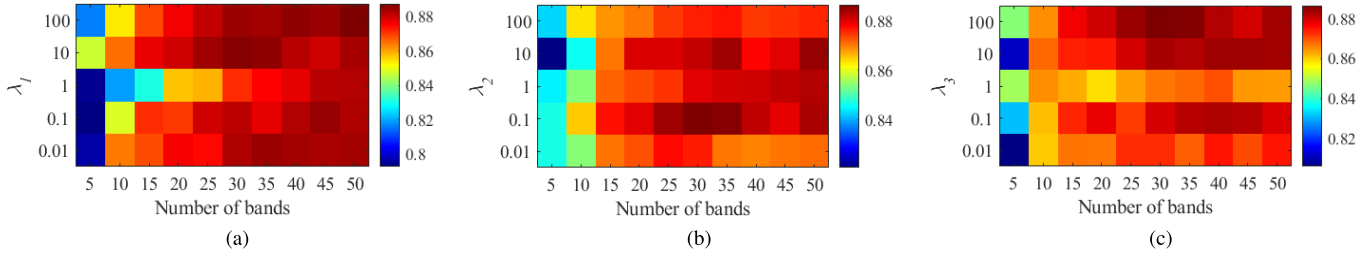


Fig. 7. Sensitivity test of parameters in terms of OA by SVM on SA dataset. (a) λ_1. (b) λ_2. (c) λ_3.

fast neighborhood grouping method for hyperspectral BS (FNGBS) [7], new neighborhood grouping normalized matched filter (NGNMF) [54], spatial and spectral structure preserved self-representation (S^4P) [55], sparse principal component analysis and adaptive multigraph learning (SPCA-AMGL) [56], DVS^3C [26], SSANet-BS [14], and capsule network-based BS for underwater hyperspectral imagery (CapUBS) [29].

2) Classifier Setting: To evaluate the classification performance of different BS methods, the support vector machine (SVM) was adopted as the classifier. For the IP, PU, and SA datasets, 10%, 1%, and 1% of the samples were randomly selected as the training set, respectively, with the remaining samples used for testing. To examine the effectiveness and stability of the methods, experiments were conducted on 10 band subsets, with the number of bands ranging from 5 to 50 in step of 5. All experiments were independently repeated five times, and the final results were reported as the average of these runs. The quality of the band subsets was assessed using three widely adopted metrics: Overall Accuracy (OA), Average Accuracy (AA), and the Kappa coefficient (Kappa).

C. Parameter Setup and Analysis

MRDSC involves four parameters: K, λ_1, λ_2 and λ_3. Specifically, K denotes the number of superpixel segmentation blocks, λ_1 controls the weight of the noise matrix regularization term, λ_2 regulates the robust coefficient matrix constraint, and λ_3 adjusts the importance of the SAC learning loss. The value of K is adaptively determined using the algorithm proposed by Jiang et al. [57], while the optimal combinations of λ_1, λ_2, and λ_3 are determined via the grid search. To prevent extreme parameter values from causing specific regularization terms to dominate or become ineffective, the search space for λ_1, λ_2, and λ_3 is set to $\{0.01, 0.1, 1, 10, 100\}$. Here, the average overall accuracy (AOA) is computed across ten band subsets, and the parameter set that achieves the highest AOA is selected as optimal. For the IP, PU, and SA datasets, the optimal parameter combinations of $\{\lambda_1, \lambda_2, \lambda_3\}$ are {1, 0.1, 0.1}, {0.1, 1, 1}, and {10, 0.1, 100}, respectively.

To evaluate the impact of each parameter on the model's performance, a univariate analysis of λ_1, λ_2, and λ_3 was conducted on the IP, PU, and SA datasets. In this analysis, only one parameter is varied at a time while the others are fixed at their optimal values, and the results are illustrated in Figs. 5–7. The results indicate that, across different datasets, the relationship between parameter variations and classification performance exhibits common patterns, while differences in specific trends and sensitivity levels are observed due to inherent data characteristics. This is attributed to the inherent variations in data characteristics. More importantly, MRDSC

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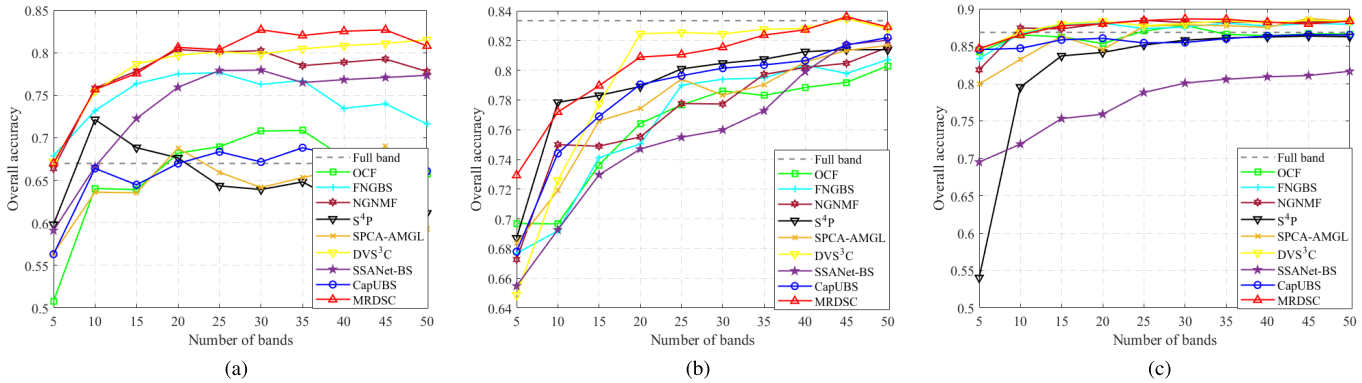


Fig. 8. OA values of SVM classifier for different BS methods. (a) IP. (b) PU. (c) SA.

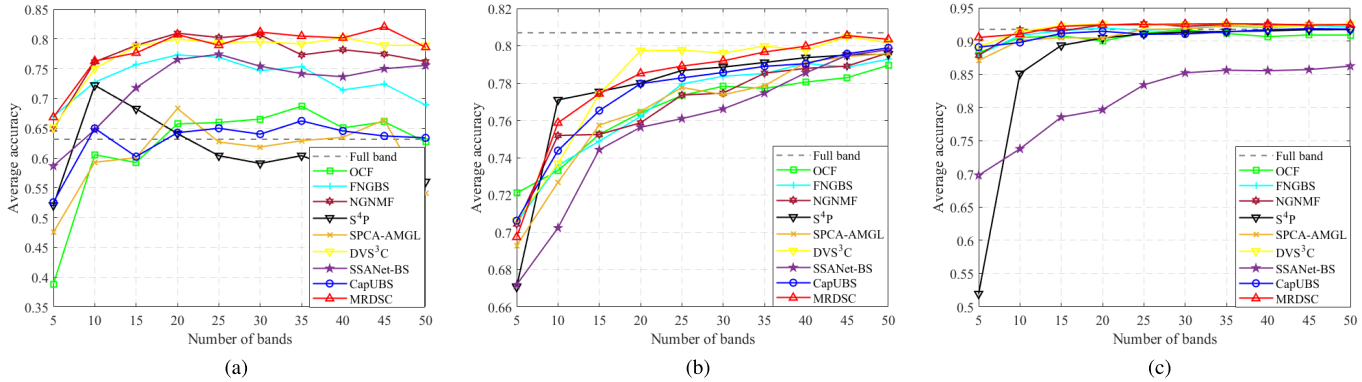


Fig. 9. AA values of SVM classifier for different BS methods. (a) IP. (b) PU. (c) SA.

demonstrates a relatively stable effective parameter range across all tested datasets; within which the model performance remains robust without noticeable degradation.

Specifically, the parameter λ_1 controls the intensity of the low-rank constraint on the coefficient matrix. An excessively small λ_1 weakens the low-rank regularization and may fail to effectively capture the underlying subspace structure, whereas an overly large value can disrupt the intrinsic manifold structure of the robust coefficient matrix, leading to feature loss. This trend is particularly evident on the IP dataset. In contrast, no pronounced performance fluctuations are observed on the PU and SA datasets, indicating that λ_1 can stably characterize the subspace structure within the tested range. The parameter λ_2 is primarily used to balance the weight between the noise term and the low-rank representation. Overall, none of the datasets exhibited high sensitivity to it. Only on the PU dataset does a noticeable performance degradation occur when λ_2 increased to 100, which may be related to the intrinsic noise level of the dataset itself. The parameter λ_3, which weights the self-alignment contrastive loss, exhibits dataset-dependent effects, likely due to differences in the degree of consistency and complementarity among multiple views. Nevertheless, within the examined parameter ranges, the overall performance of the model remains stable across all datasets.

In summary, although the optimal parameters vary slightly across datasets due to their inherent characteristics, MRDSC consistently exhibits a clear and stable effective parameter range on all tested datasets. Satisfactory performance can be achieved without fine-tuning the parameters for each individual dataset. These results demonstrate the robustness of the proposed method with respect to parameter settings and its strong generalization capability across different datasets.

D. Results Analysis

Figs. 8–10 present the OA, AA, and Kappa values of different methods on ten band subsets across three datasets. As illustrated in Fig. 8, OCF and SSANet-BS do not demonstrate clear advantages. S^{4}P, SPCA-AMGL, and CapUBS achieve competitive results on specific datasets but suffer from large performance fluctuations. For instance, S^{4}P and CapUBS achieves satisfactory accuracy on the PU dataset, but performs poorly on the IP and SA datasets. In contrast, FNGBS, NGNMF and DVS^{3}C show relatively stable performance across the three datasets, though they still fail to achieve the best results consistently. The proposed MRDSC performs excellently on all datasets and achieves the best results on all of them. On the IP dataset, MRDSC maintains a leading advantage on all band subsets, with its superiority becoming more evident when the number of bands exceeds 25. On the PU dataset, while DVS^{3}C outperforms MRDSC in the intermediate band range, its performance at low band numbers is poor. In contrast, MRDSC maintains highly consistent performance across the entire spectral range, demonstrating a stronger comprehensive advantage in robustness. On the SA dataset, although MRDSC does not yield the highest OA on

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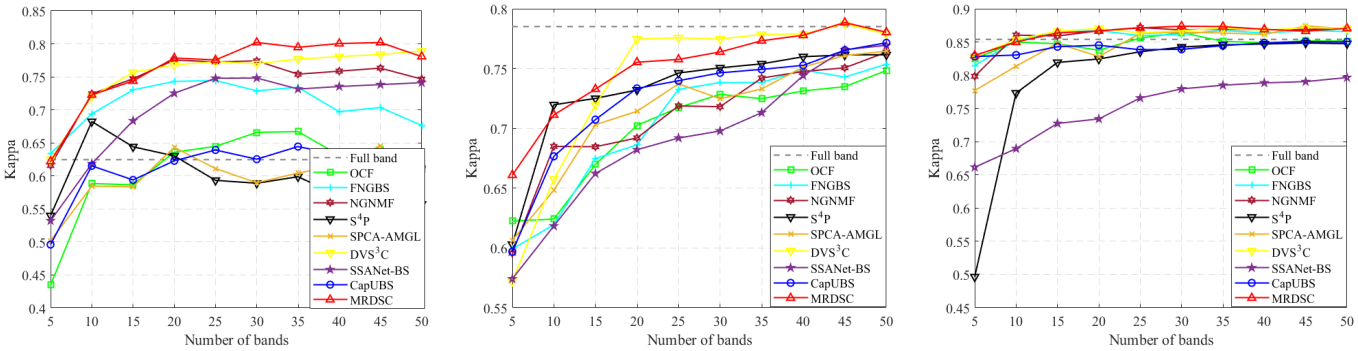


Fig. 10. Kappa values of SVM classifier for different BS methods. (a) IP. (b) PU. (c) SA.

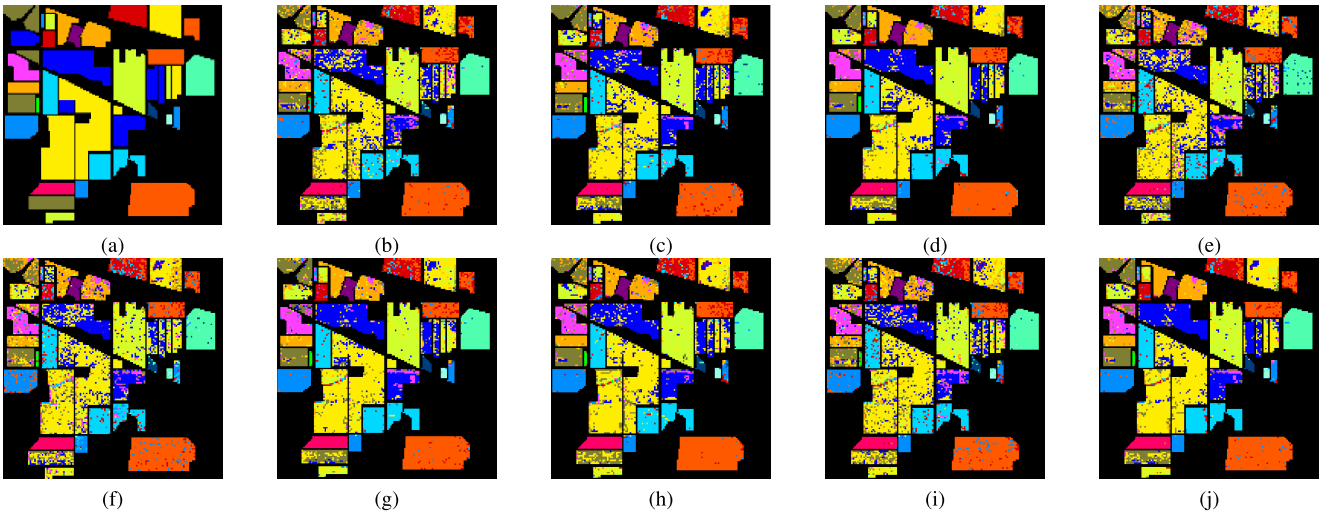


Fig. 11. Classification maps of various BS methods under 30 bands for IP dataset. (a) Ground truth. (b) OCF (70.84%). (c) FNGBS (77.05%). (d) NGNMF (80.59%). (e) S^4P (65.57%). (f) SPCA-AMGL (65.47%). (g) DVS^3C (80.85%). (h) SSANet-BS (78.21%). (i) CapUBS (68.26%). (j) MRDSC (83.01%).

a few individual band subsets, it still ranks among the top methods.

OA reflects the overall classification accuracy, whereas AA and Kappa emphasize the balance of accuracy across different classes. As shown in Figs. 9 and 10, on the IP dataset, MRDSC is slightly surpassed by NGNMF when the number of bands is 15 and 25. For the PU dataset, while MRDSC is slightly outperformed by DVS^3C in certain band subsets, it maintains highly competitive performance over all other comparison methods across the remaining subsets. On the SA dataset, it still performs robustly. In summary, the band subset selected by MRDSC achieves good classification performance on both large and small classes.

To intuitively demonstrate the performance of different BS methods, Figs. 11–13, respectively, show the classification maps and their corresponding OA values on the three datasets with 30 bands. It can be seen that the proposed MRDSC achieves the highest classification accuracies, with OA values of 83.01%, 83.91%, and 88.83%, respectively. For the IP dataset, the performance of all methods is relatively low, and only NGNMF, DVS^3C, and MRDSC achieve OA values above 80%. Although NGNMF and DVS^3C surpasses the 80% threshold, it still lags behind MRDSC. For the PU dataset, MRDSC leads the second-ranked DVS^3C by 0.33%. Meanwhile, for the SA dataset, the results of different BS methods are comparable, yet the proposed method MRDSC consistently attains the best performance.

Furthermore, to evaluate the overall performance of all methods, Table I shows the average OA, average AA, and average Kappa values for different BS methods on the three datasets. These metrics are averaged over 10 band subsets, ranging from 5 to 50 with a step of 5. The optimal and suboptimal values are highlighted in red bold and black bold, respectively. As shown in the table, MRDSC achieves the highest AOA values on the three datasets, with 79.19%, 80.42%, and 87.71%, respectively. The trends of the AA and Kappa values are highly consistent with the trend of the OA values, validating the effectiveness of the proposed method. It is notably that DVS^3C attained the suboptimal AOA values on the three datasets. This result further substantiating the superiority of multiview strategies for improving the discriminability of the selected band subsets.

V. Discussion

A. Band Quality

To visually evaluate the quality of the band subsets obtained by different BS methods, Fig. 14 illustrates the distribution of selected bands and their corresponding SE values on the

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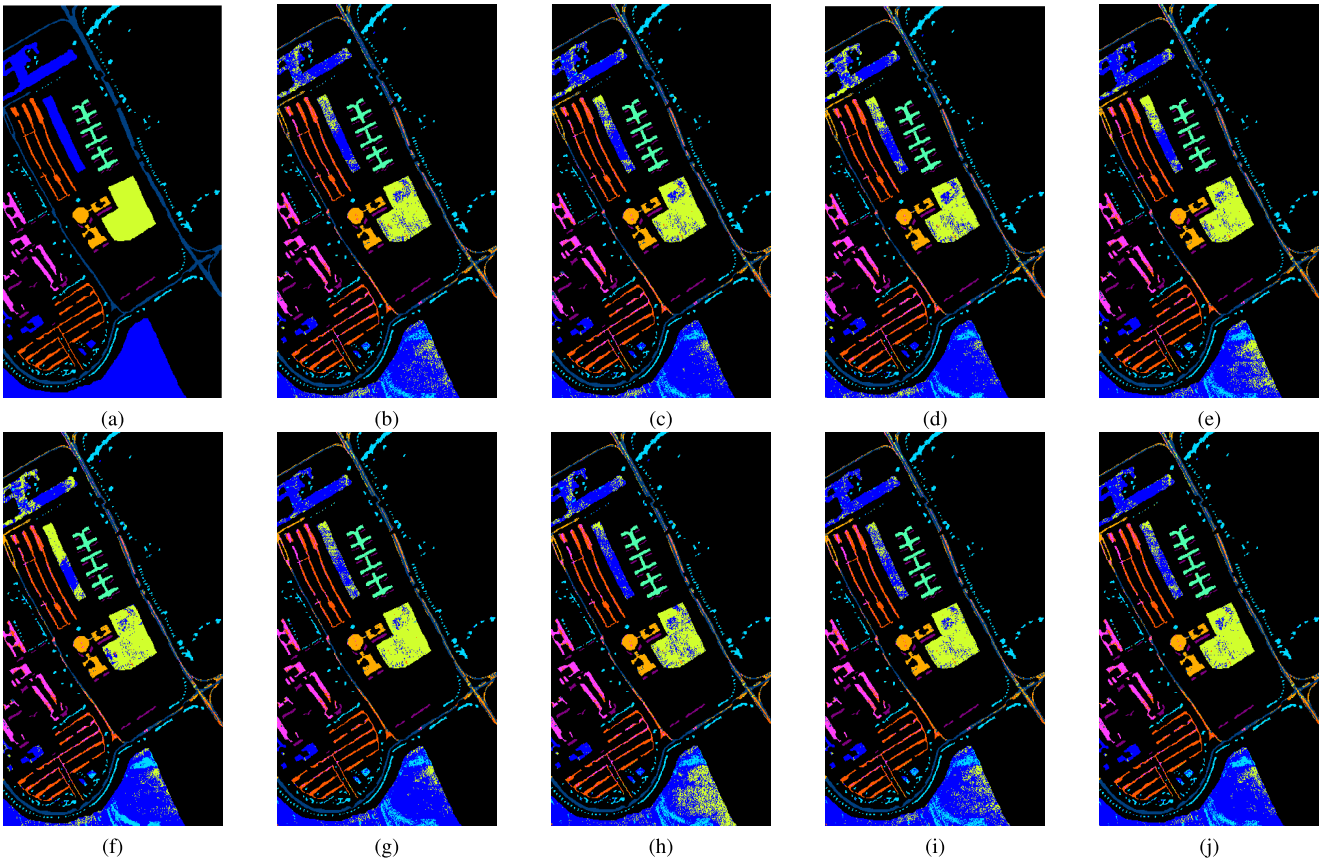


Fig. 12. Classification maps of various BS methods under 30 bands for PU dataset. (a) Ground truth. (b) OCF (81.49%). (c) FNGBS (83.50%). (d) NGNMF (81.65%). (e) S^4P (80.90%). (f) SPCA-AMGL (79.12%). (g) DVS^3C (83.58%). (h) SSANet-BS (77.21%). (i) CapUBS (81.12%). (j) MRDSC (83.91%).

TABLE I
CLASSIFICATION RESULT (%) FOR DIFFERENT BS METHODS ON THREE DATASETS

Methods	IP			PU			SA		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
OCF	65.95	61.94	61.05	76.23	76.52	70.04	86.34	90.60	84.83
FNGBS	74.47	73.24	70.86	76.47	76.72	70.34	87.25	91.45	85.83
NGNMF	77.51	77.09	74.32	77.01	76.74	70.98	87.42	91.81	86.02
S^4P	64.49	60.52	59.49	78.91	77.47	73.14	81.73	86.56	79.80
SPCA-AMGL	64.24	60.66	59.15	77.46	76.53	71.44	86.18	91.11	84.63
DVS^3C	**78.48**	**77.50**	**75.43**	**79.45**	**78.12**	**73.98**	**87.67**	**91.98**	**86.30**
SSANet-BS	73.75	72.29	70.02	75.48	75.55	69.19	77.56	81.34	75.18
CapUBS	65.39	62.89	61.07	78.28	77.37	72.40	85.75	91.02	84.18
MRDSC	**79.19**	**78.28**	**76.23**	**80.42**	**78.02**	**75.02**	**87.71**	**92.13**	**86.34**

IP dataset when the number of bands is 30. This figure comprises two subplots: the first subplot displays the positional distribution of the 30 bands selected by each BS method, where points along the line represent their index positions across the full spectral range. The effectiveness of the spectral band distribution is crucial. Generally, strong correlations often exist between adjacent and consecutive bands. Conversely, a more dispersed distribution of bands across the spectral range is more conducive to capturing diverse spectral features. The second subplot further presents the SE value of each band, quantifying its information content. Higher entropy value indicates greater information richness within the band.

As shown in Fig. 14, the band distribution of SPCA-AMGL is relatively broad. However, in the IP dataset, the bands [104–108] and [150–163] are known as water absorption bands with significant noise [58]. It shows a higher distribution density in these noisy band regions, and their performance is inevitably compromised by this low-quality information. NGNMF, OCF, and CapUBS have a wide distribution, but their selected bands include low-entropy regions. This leads to insufficient information in their band subsets, making it difficult to provide adequate discriminative information to support the subsequent land cover identification task. DVS^3C successfully targets high-entropy intervals but its selected bands are densely grouped in the first half of the spectrum while neglecting the information contained in the higher spectral range. It may negatively affect the downstream classification task. The bands selected by FNGBS, SSANet-BS,

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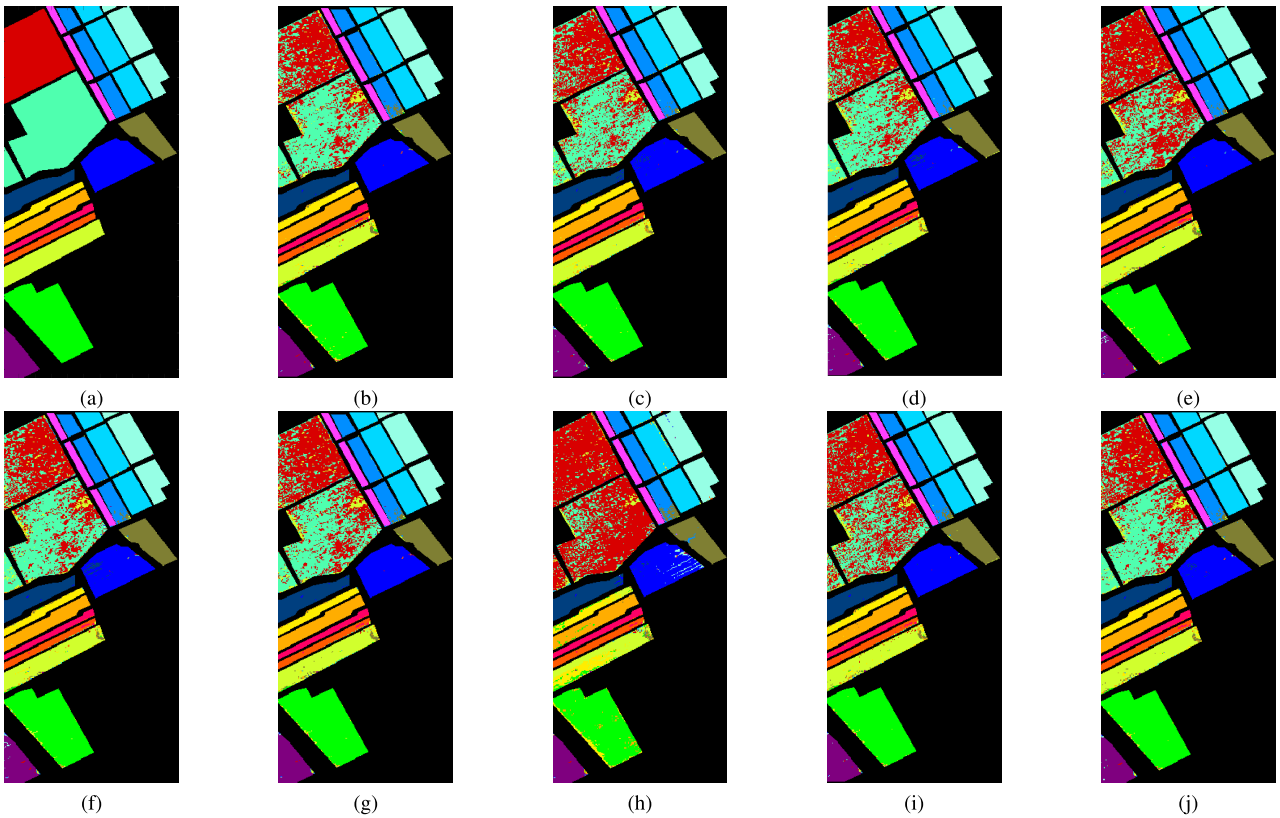


Fig. 13. Classification maps of various BS methods under 30 bands for SA dataset. (a) Ground truth. (b) OCF (88.70%). (c) FNGBS (87.77%). (d) NGNMF (88.02%). (e) S^4P (85.33%). (f) SPCA-AMGL (88.16%). (g) DVS^3C (88.12%). (h) SSANet-BS (76.16%). (i) CapUBS (85.05%). (j) MRDSC (88.83%).

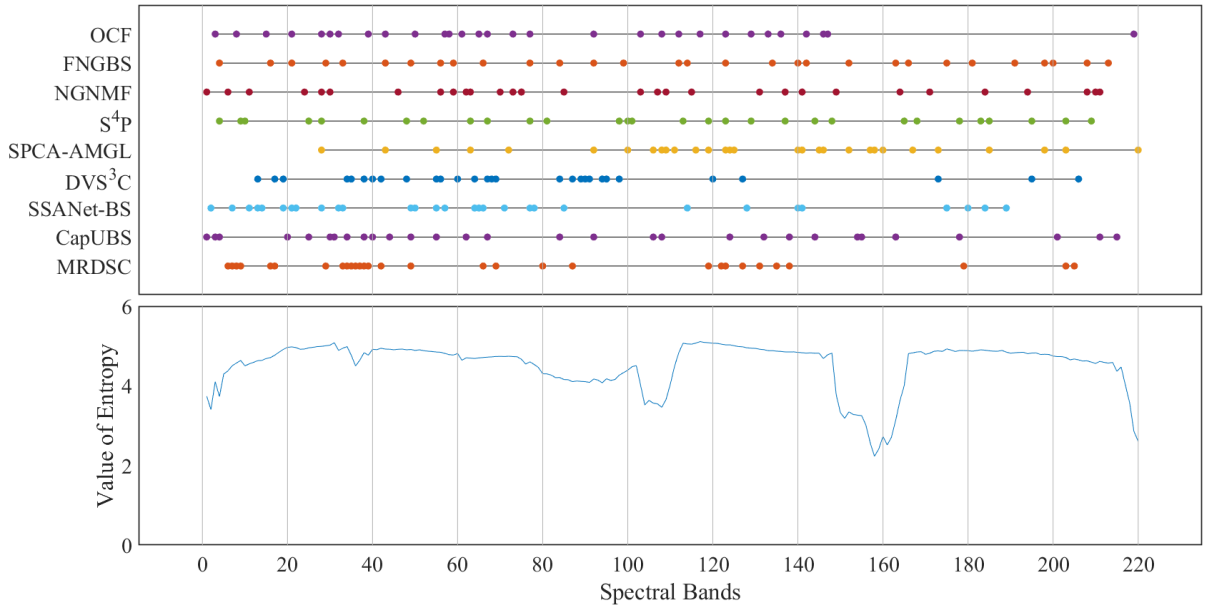


Fig. 14. 30 bands selected by different BS methods for IP dataset. The above subgraph represents the index of 30 selected bands, and the below subgraph represents the entropy value of all bands.

and MRDSC not only cover a wide spectral range but also effectively avoid low-entropy regions. However, as further validated by the OA values presented in Fig. 8, while FNGBS and SSANet-BS perform commendably in spectral dispersion and low-entropy avoidance, the band subsets selected by MRDSC demonstrate superior capability in enhancing discriminability among land cover classes. This further demonstrates the advantages of MRDSC in suppressing redundancy and avoiding noisy and low-information bands, proving the effectiveness of MRDSC.

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TABLE II
AOA VALUES (%) OF FOUR VIEW CONFIGURATIONS ON THREE DATASETS

Methods	View1	View2	View3	IP	PU	SA
(a)	✓			77.91	79.61	87.08
(b)	✓	✓		79.16	80.07	87.65
(c)	✓		✓	79.06	80.06	87.58
(d)	✓	✓	✓	**79.19**	**80.42**	**87.71**

TABLE III
AOA VALUES (%) OF FOUR MODELS FOR THREE DATASETS

Methods	ML	RPCA	SAC	IP	PU	SA
(a)				76.10	77.94	87.05
(b)	✓			77.87	78.06	87.19
(c)	✓	✓		78.52	79.13	87.21
(d)	✓	✓	✓	**79.19**	**80.42**	**87.71**

B. Ablation Study on Single-View and Multiview Models

To investigate the contribution of each view in MRDSC, four configurations were designed based on the spectral view (view1), spatial view (view2), and texture view (view3). Specifically, four configurations are evaluated: 1) a single-view setting using only the spectral view; 2) a dual-view setting combining spectral and spatial views; 3) a dual-view setting combining spectral and texture views; and 4) a three-view setting integrating spectral, spatial, and texture views. Table II summarizes the AOA values for these configurations on the three datasets, with the best highlighted in bold red.

The experimental results clearly show that the single-view configuration using only the spectral view (1) yields the lowest AOA, indicating that spectral features alone are insufficient to characterize the complex interband relationships. With the introduction of either the spatial or texture view, the model performance improves significantly. In particular, the spatial view effectively enhances the modeling of global and local spatial structural relationships among bands, while the texture view further refines feature representations by complementing local structural and detailed information. Ultimately, the three-view joint modeling achieves the highest AOA, convincingly demonstrating that spectral, spatial, and texture views can collaboratively construct a more comprehensive band similarity representation. Overall, all three views play indispensable roles in improving BS performance.

C. Ablation Study on Different Modules in MRDSC

To further demonstrate the effectiveness of the multilevel DSC (ML), RPCA, and SAC loss modules in MRDSC, this section designed four models.

1) *Base Model:* A DSC-based BS model that only uses a single-level feature representation.
2) *Base Model + ML:* Introduces multilevel FE in DSC.
3) *Base Model + ML + RPCA*: Adds RPCA decomposition to (2).
4) *MRDSC*: Incorporates the SAC loss into (3). The experimental results of these four models for the three datasets are presented in Table III. The optimal values are highlighted in bold red.

TABLE IV
RUNTIME (S) FOR VARIOUS BS METHODS UNDER 30 BANDS FOR THREE DATASETS

Methods	IP	PU	SA
OCF	0.15	0.41	0.50
FNGBS	0.16	0.36	0.46
NGNMF	0.25	0.51	0.62
S^4P	0.78	4.54	3.61
SPCA-AMGL	1.44	0.94	1.10
DVS^3C	4.42	9.45	17.62
SSANet-BS	37.26	197.11	206.36
CapUBS	$>$ 10h	$>$ 10h	$>$ 10h
MRDSC	69.58	58.53	98.60

TABLE V
COMPLEX METRICS OF DL-BASED BS METHODS

Methods	Datasets	FLOPs (G)	MACs (G)	Parameters (K)
SSANet-BS	IP	1011.69	505.85	905.23
	PU	4933.64	2466.82	557.16
	SA	5851.34	2925.67	921.49
CapUBS	IP	3903.99	1952.00	1456353.27
	PU	9270.35	4635.18	299379.56
	SA	22990.67	11495.34	1513015.54
MRDSC	IP	14.54	7.27	212.77
	PU	68.34	34.17	92.41
	SA	79.44	39.72	329.81

Model (1) performs worse than the other three models on all three datasets, indicating that single-level latent features are insufficient to fully exploit the intrinsic correlations among bands within each view. Model (2) shows a significant improvement over model (1). This demonstrates that multilevel latent features better leveraged the local–global relationships of each view to obtain a more comprehensive coefficient matrix. Compared to model (2), model (3) is further improved. This benefits from RPCA, which decomposes the coefficient matrix into a cleaner matrix $\mathbf{L}$, effectively suppressing outlier interference and enhancing the cluster discriminability. In addition, method (4), the proposed MRDSC, achieves the optimal performance on all three datasets with a significant advantage over the suboptimal methods. This indicates that the SAC loss can effectively fuse the complementary and consistent information from each view, improving the BS performance. Overall, these experiments confirm that the combination of multilevel DSC, RPCA decomposition, and the SAC loss can significantly enhance the ability of SC to represent band correlations, thereby selecting information-rich and discriminative band subsets.

D. Analysis of Computational Efficiency

To comprehensively evaluate the computational efficiency of MRDSC, we conducted a systematic analysis of all comparison methods. Specifically, to intuitively assess the practical processing capability of each method, Table IV reports the runtime (s) of all methods with 30 selected bands. All experiments

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were conducted on a unified hardware platform equipped with an Intel i7-13700 CPU, an NVIDIA RTX 4090 GPU, and 32GB of RAM. The six machine learning-based methods, OCF, FNGBS, NGNMF, S^4P, SPCA-AMGL, and DVS^3C were implemented in MATLAB, while the three DL models, SSANet-BS, CapUBS, and MRDSC, were implemented in Python and accelerated on the GPU. In addition, Table V compares key complexity metrics for the DL models, including FLOPs, MACs, and the number of parameters.

Table IV shows that machine learning-based methods do not require complex model training and thus have an inherent advantage in runtime. However, their FE capability is limited, resulting in substantially lower classification performance of the selected band subsets compared with MRDSC. In contrast, DL-based methods significantly enhance the modeling of discriminative features in HSIs during the training phase by stacking deep convolutions, attention mechanisms, or fully connected layers. This, however, introduces a large number of floating-point operations and parameters, leading to significantly higher overall runtime. Even with GPU acceleration, their computational cost remains much greater than that of traditional machine learning-based methods, reflecting the tradeoff of increased resource consumption for performance gains. MRDSC achieves relatively low runtime across multiple benchmark datasets, outperforming existing DL models such as SSANet-BS and CapUBS, while maintaining superior classification accuracy. This advantage is primarily attributed to MRDSC's lightweight convolutional-transposed convolutional symmetric architecture and the efficient band-dimension self-representation mechanism, which effectively simplifies the computational path while ensuring the model's ability to extract deep discriminative features.

Furthermore, as shown in Table V, MRDSC achieves FLOPs of 14.54, 68.34, and 79.44 G on the three datasets, respectively, demonstrating a clear order-of-magnitude reduction in computational cost compared with SSANet-BS and CapUBS. Moreover, MRDSC exhibits significantly lower MACs, indicating higher computational efficiency. In terms of model compactness, MRDSC requires only 212.77, 92.41, and 329.81 K parameters on the three datasets, respectively, which is substantially fewer than those of SSANet-BS and CapUBS. Overall, MRDSC achieves superior accuracy with substantially lower computational and storage overhead than SSANet-BS and CapUBS. This strikes a favorable balance between model accuracy and resource consumption, highlighting its feasibility and practicality for real-world hyperspectral applications.

E. Tradeoff Between Number of Views and Computational Overhead

To clarify the necessity of the three-view strategy introduced in MRDSC and the resulting additional computational overhead, we conduct an in-depth analysis from three perspectives: computational complexity, application rationality, and methodological contribution.

1) Complexity Perspective: The complexity increase due to view expansion is linear, controllable, and expected. In practice, expanding from two views to three increases both computation load and the number of parameters by approximately 1.5 times. This linear relationship aligns with expectations for adding additional views, indicating that the complexity growth in the proposed multiview framework is both predictable and controllable, without introducing excessive redundancy or nonlinear increases in computational cost.

2) Application Perspective: Hyperspectral BS is typically conducted as an offline preprocessing step prior to classification or detection tasks, requiring only a single execution and thus having no direct impact on the training or inference efficiency of subsequent models. As a result, introducing a moderate increase in computational cost at this stage to improve the quality of the selected feature subset is acceptable in practical applications. Given that existing state-of-the-art methods have already achieved strong performance, even limited but consistent improvements-particularly in terms of stability, robustness, and adaptability to complex scenarios-remain valuable. Therefore, the tradeoff of "time for performance" in offline BS is reasonable and meaningful in specific scenarios.

3) Methodological Perspective: The core contribution of this work is not merely increasing the number of views, but rather presenting the first attempt to incorporate multiview modeling into a DSC-BS framework. Most existing methods primarily focus on spectral information, while the joint exploitation of spatial structure or texture priors remains limited. By integrating multiview representations with DSC mechanisms, the proposed MRDSC enables the collaborative modeling of complementary information from different modalities within a unified framework. Although the performance gain of using three views over two views may appear marginal on standard datasets, its methodological value and extension potential are noteworthy. Moreover, the proposed framework is theoretically extensible and not restricted to the specific views adopted in this study, providing a flexible foundation for incorporating additional view information in future work.

In summary, the computational overhead introduced by the three-view strategy is linear and controllable. In the context of offline BS, trading a moderate increase in computation for improved performance and robustness is reasonable. More importantly, this work explores a unified modeling framework that integrates multiview information with DSC for hyperspectral BS, offering valuable insights for future research on multisource information fusion.

VI. Conclusion

This article proposes a novel hyperspectral BS model, MRDSC. The model first constructs a spatial view and a texture view based on the spectral view. This aims to mine the joint spectral, spatial, and textural information of the HSI from multiple perspectives. Subsequently, it employs a multilevel robust DSC framework to explore fine linear-nonlinear relationships among bands across views. Meanwhile, SAC learning is used to enhance the consistency of positive samples and improve the discriminability of negative samples across views, thereby effectively fusing the complementary and consistent information among the views. Experimental results on three real datasets demonstrate that the proposed

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MRDSC is stable, effective, and superior to other comparison methods.

Although MRDSC explores the impact of the spatial view and the texture view on BS, many potential feature views remain unexplored. Furthermore, designing more effective view fusion strategies still poses challenges. Future work will primarily focus on exploring more valuable view criteria and designing more efficient view fusion strategies. This aims to construct a more comprehensive consistent matrix representation and obtain better performance of BS.

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